

Breaking change: BinaryFormatter serialization APIs produce compiler errors

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Content

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BinaryFormatter serialization APIs produce compiler errors

Summarize this article for me

As part of the BinaryFormatter [long-term deprecation plan](#), we continue to cull BinaryFormatter functionality from our libraries and to wean developers off of the type. Starting in .NET 7, calls to the following APIs produce compile-time errors across all C# and Visual Basic project types:

- [System.Exception.SerializeObjectState](#) event
- [BinaryFormatter.Serialize](#) method
- [BinaryFormatter.Deserialize](#) method
- [Formatter.Serialize\(Stream, Object\)](#) method
- [Formatter.Deserialize\(Stream\)](#) method
- [IFormatter.Serialize\(Stream, Object\)](#) method
- [IFormatter.Deserialize\(Stream\)](#) method

Previous behavior

Since .NET 5, using the affected `Serialize` and `Deserialize` methods produced a compiler *warning* with ID `SYSLIB0011`. For more information, see [BinaryFormatter serialization methods are obsolete and prohibited in ASP.NET apps \(.NET 5\)](#).

Using the `Exception.SerializeObjectState` event did not produce an error.

New behavior

Starting in .NET 7, using any of the [affected APIs]() in code produces a compiler *error* with the same ID, `SYSLIB0011`. Your project will be affected if it meets all of the following criteria:

- It's a C# or Visual Basic project.
- It targets `net7.0` or higher.
- It directly calls one of the [affected APIs]().
- It's not already suppressing the `SYSLIB0011` warning code.

Version introduced

.NET 7

Type of breaking change

This change can affect [source compatibility](#).

Reason for change

As part of the BinaryFormatter [long-term deprecation plan](#), we continue to cull `BinaryFormatter` functionality from our libraries and to wean developers off of the type.

The best course of action is to migrate away from `BinaryFormatter` due to its security and reliability flaws. `BinaryFormatter` may be removed from .NET in a future release. The .NET libraries team has already taken a stance that recent types such as `System.Half` and `System.DateOnly` won't be compatible with `BinaryFormatter`.

If you must suppress the errors, you can do so by following the guidelines in the [original obsolescence article](#). You can also disable the error project-wide by setting a project property that converts the error back to a warning (to match the .NET 5/6 behavior).

Warning

Setting this property might change host behavior. See `[<EnableUnsafeBinaryFormatterSerialization> property]()`.

```
<PropertyGroup>
    ...

    <EnableUnsafeBinaryFormatterSerialization>true</EnableUnsafeBinaryFormatterSerialization>
</PropertyGroup>
```

Note

If your project compiles with "warnings as errors" enabled, compilation will still fail. (This matches the behavior that shipped in the .NET 5 and .NET 6 SDKs.) If that's the case, you'll still need to suppress the `SYSLIB0011` warning in source or in your project file's `<NoWarn>` element.

`<EnableUnsafeBinaryFormatterSerialization>` property

The `<EnableUnsafeBinaryFormatterSerialization>` property was introduced in .NET 5. With .NET 7, the behavior of this switch has changed to control *both compilation and host* runtime behavior. The meaning of this switch differs based on the project type, as described in the following table.

Type of project	Property set to <code>true</code>	Property set to <code>false</code>	Property omitted
Library/shared component1	The affected APIs are obsolete as warning. Compilation will succeed unless you have "warnings as errors" enabled for your application or you've suppressed the <code>SYSLIB0011</code> warning code.	The affected APIs are obsolete as error, and calls from your code to those APIs will fail at compile time unless the error is suppressed. (Same as for <code>false</code> .)	Blazor and MAUI apps2
	Calls to <code>BinaryFormatter</code> will fail at runtime.	Calls to <code>BinaryFormatter</code> will fail at runtime.	Calls to <code>BinaryFormatter</code> will fail at runtime.
ASP.NET app	The affected APIs are obsolete as warning. Compilation will succeed unless you have "warnings as errors" enabled for your application or you've suppressed the <code>SYSLIB0011</code> warning code. The runtime will <i>allow</i> calls to <code>BinaryFormatter</code> , regardless of whether the call originates from your code or from a dependency that you consume.	The affected APIs are obsolete as error, and calls from your code to those APIs will fail at compile time unless the error is suppressed. The runtime will <i>forbid</i> calls to <code>BinaryFormatter</code> , regardless of whether the call originates from your code or from a dependency that you consume. (Same as for <code>false</code> .)	Desktop apps and all other app types
	The affected APIs are obsolete as warning. Compilation will succeed unless you have "warnings as errors" enabled for your application or you've suppressed the <code>SYSLIB0011</code> warning code. The runtime will <i>allow</i> calls to <code>BinaryFormatter</code> , regardless of whether the call originates from your code or from a dependency that you consume.	The affected APIs are obsolete as error, and calls from your code to those APIs will fail at compile time unless the error is suppressed. The runtime will <i>forbid</i> calls to <code>BinaryFormatter</code> , regardless of whether the call originates from your code or from a dependency that you consume.	
	The affected APIs are obsolete as error, and calls from your code to those APIs will fail at compile time unless the error is suppressed. The runtime will <i>allow</i> calls to <code>BinaryFormatter</code> , regardless of whether the call originates from your code or from a dependency that you consume.		

1Runtime policy is controlled by the app host. Calls into `BinaryFormatter` might still fail at runtime even if `<EnableUnsafeBinaryFormatterSerialization>` is set to `true` within your library's project file. Libraries can't override the app host's runtime policy.

2The Blazor and MAUI runtimes forbid calls to `BinaryFormatter`. Regardless of any value you set for `<EnableUnsafeBinaryFormatterSerialization>`, the calls will fail at runtime. Don't call these APIs from Blazor or MAUI applications or from libraries intended to be consumed by Blazor or MAUI apps.

Affected APIs

- [System.Exception.SerializeObjectState](#)
- [System.Runtime.Serialization.Formatters.Binary.BinaryFormatter.Serialize](#)
- [System.Runtime.Serialization.Formatters.Binary.BinaryFormatter.Deserialize](#)
- [System.Runtime.Serialization.Formatter.Serialize\(Stream, Object\)](#)
- [System.Runtime.Serialization.Formatter.Deserialize\(Stream\)](#)
- [System.Runtime.Serialization.IFormatter.Serialize\(Stream, Object\)](#)
- [System.Runtime.Serialization.IFormatter.Deserialize\(Stream\)](#)

See also

- [dotnet/runtime issue 72132](#)
- [BinaryFormatter serialization methods are obsolete \(.NET 5\)](#)
- [SerializationFormat.Binary is obsolete \(.NET 7\)](#)
- [BinaryFormatter disabled across most project types \(.NET 8\)](#)

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